THE PERIODIC TABLE

INTRODUCTION

The periodic table was first published by Mendeleev in 1869. The modern form of the table is the one you are familiar with. The Periodic Table places the elements in order of Atomic Number, in Groups of similar chemical properties. You need to know the names and symbols of the first 20 elements and a few more.

EXERCISES

1. Place the following elements on the periodic table: (Use a black pen)
   METALS : Li, Be, Na, Mg, Al, K, Ca, Fe, Cu, Zn, Pb, Ag, Sn, Ba, Au,
   NON METALS : C, N, O, F, H, He, P, S , Cl, Br, I, Ne, Ar, Kr,
   METALLOIDS : Si, B

2. On the chart, colour in the three areas of metals, nonmetals and metalloids in different colours.

3. Using a completed periodic table, place the atomic number (number of protons) for each element on your sheet. Put it at the bottom of your periodic table regardless of where it is on your completed table.

4. At the top of each box, write the name of your element in small writing.

5. Number the groups across from 1 to 18. Label Group 1 – Alkali metals, Group 2 – Alkaline Earth metals, Gp 17 – Halogens, Gp 18 – Noble gases, Gps 3-12 – Transition metals.

QUESTIONS

1. Which elements have symbols that do not begin with the element's name? Can you find the origin of these names? HINT: The symbols are derived from Latin.

2. Some groups have special names. From your table, list the elements from the following groups :- Noble gases, Halogens, Alkali Metals.

3. A dark staircase – shaped line separates the metals form the nonmetals. For the first 80 elements, find the number of metals and nonmetals. Calculate the percentage of metals in the first 80 elements.

4. How many symbols can you make using your full name? (Use the lab's periodic table also). Try to name all the symbols you find.